ARNOB ADHIKARY

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EDUCATION

Feb. 2018 – Aug. 2023
Khulna, Bangladesh

B.Sc. in Building Engineering and Construction Management (BECM)
Khulna University of Engineering & Technology (KUET)
CGPA: 3.28 (Out of 4.00)
GPA in the last 60 credits: 3.65 (Out of 4.00)
Major: Architectural Engineering

RESEARCH EXPERIENCE

Undergraduate Thesis: Evaluation of Thermal Performance of a Wall Integrated Evaporative Cooling System with Natural Ventilation in Warm Humid Climate
Supervisor: Md Shahriar Hossain

- Conducted an evaluation of a wall-integrated evaporative cooling system's efficacy in a humid climate. Employed CFD with ANSYS FLUENT to simulate airflow and heat transfer under various humidity and velocity conditions. Validated simulation findings through the design, construction, and experimental testing of physical micro-models. Results established a critical inverse relationship between cooling efficiency and ambient relative humidity, highlighting proficiency in CFD analysis, experimental validation, and data interpretation.

Independent Research (In progress): Multimodal Asset Identification and Semantic Enrichment of As-Built BIM for Facility Management.
Self-directed; developing in collaboration with two colleagues from KUET's BECM

- Developing a two-stage AI framework for facility management: Stage 1 identifies building assets from 360° scan imagery and equipment schedules; Stage 2 retrieves and populates verified, evidence-grounded semantic asset data into the as-built BIM model.
- Built and validated an initial 2D-to-3D asset localization pipeline on real laser-scan data (four scan stations, E57 format with 360° panoramic imagery) using Python, YOLO, and Open3D.
- Currently extending the approach with trained segmentation models and incorporating equipment schedules as a structured prior for asset identification in Stage 1.

RESEARCH INTERESTS

Building Information Modeling (BIM), LiDAR and Photogrammetry in Built Environment, Digital Twin, AR & VR in construction, Facility Management, Construction Safety, Construction Automation.

TECHNICAL SKILLS

BIM & CAD	**:** Revit, BlenderBIM, Stabicad, Navisworks, AutoCAD, Microstation
Programming & Scripting	**:** Python, Dynamo, FME, AutoLISP, Javascript, HTML & CSS
Photogrammetry & LiDAR	**:** Terrasolid Suite, Agisoft Metashape, Trimble Inpho
Structural Analysis	**:** ETABS, Autodesk Robot
Documentation	**:** MS Office, LaTeX, Adobe InDesign

PROFESSIONAL EXPERIENCE

Netherlands/Bangladesh **HawarIT Limited**
Dec. 2023 – Present BIM Coordinator | Full Time

- Managed the successful execution of over 170 projects in Scan-to-BIM and Digital Terrain Modeling, leading a multidisciplinary team of 20+ specialists and managing the entire project lifecycle from client communication and proposal development to final delivery.
- Standardized a comprehensive Scan-to-BIM workflow, converting complex static LiDAR data into multidisciplinary BIM models (LOD 300-400) in Revit for over 100 industrial and infrastructure clients.
- Initiated workflow automation by developing custom scripts using Dynamo, IfcOpenShell (Python), FME (Feature Manipulation Engine) reducing manual data processing and modeling hours.
- Processed over 500,000+ sq. m. of MLS data to deliver high-resolution DTM and planimetric maps for Amsterdam municipalities, performing precise GCP-based matching and point cloud classification to achieve a final geometric accuracy of <2 cm RMSE using Terrasolid, Microstation, FME.
- Designed a photogrammetry pipeline in Agisoft Metashape to process UAV imagery, producing accurately georeferenced orthomosaics with a 2 cm GSD for a land ministry project in Bangladesh.
- Directed the digitization and data management of critical technical drawings for high-profile clients, including the European Union's Joint Research Centre (JRC), Stitching Yuverta and Gemeente Utrecht by developing survey and digitization workflow to ensure data integrity.
- Secured projects valued at over €280,000 annually by authoring compelling technical proposals and managing key client relationships.

Dhaka, Bangladesh **ideaMods**
Mar. 2020 – Dec. 2020 Online Operation and Marketing Coordinator | Part Time

- Developed and deployed an automated order and inventory management system using Google Sheets and Google Apps Script.
- Managed all client-facing social media channels, handling customer inquiries, processing custom furniture orders, and executing targeted digital marketing campaigns to boost engagement and sales during the COVID-19 pandemic.

ACCOLADES

2020 **1st Runner Up in CAD Contest** in BUILTECH FEST 4.0, KUET.
2020 **1st Runner Up in Visual Rendition** in BUILTECH FEST 4.0, KUET
2019 **Awarded Technical Scholarship** during the undergraduate first year for achieving a place within the top half of the merit rankings
2015 **Bangladesh Government Scholarship** for outstanding performance in Secondary School Certificate exam

LEADERSHIP & TEACHING EXPERIENCE

2020 – 2021 **Treasurer of BECM Technical Club, KUET (BTECK)**
Led technical workshops in Revit, AutoCAD, and SketchUp for 50+ students while simultaneously managing all club finances and event budgets as Treasurer.

2019 – 2023 **Private Tutor**
Mentored more than 20 senior-year high school (12th Grade) students in Mathematics and Physics in 5 years.

INDEPENDENT PROJECT

Dec. 2022 – Apr. 2023

Architectural Design of a 5-Storey Residential Building

Designed the complete architectural layout of a 5-storey residential building independently. Produced 3D models and photorealistic renders. Finally, delivered full working drawings for construction.

TRAINING & CERTIFICATION

July 2026

Survey Training

4-Day Comprehensive Geospatial Survey Training using UAV, RTK GNSS, Total Station & GPS with Data Processing.

REFERENCES

Md. Shahriar Hossain
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